

DIMITRIS OIKONOMOU

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EDUCATION

Johns Hopkins University <i>PhD in Computer Science</i> Advisor: Dr. Nicolas Loizou	Baltimore, USA 08/2023 - Present
National Technical University of Athens <i>MSc in Data Science and Machine Learning</i> Thesis: Efficient Learning Algorithms of Time Varying Ranking Distributions, Advisor: Dr. Dimitris Fotakis	Athens, Greece GPA: 9.8/10
University of Göttingen <i>MSc in Mathematics</i> Thesis: Convergence Results for Entropic Transfer Operators, Advisor: Dr. Bernhard Schmitzer	Göttingen, Germany GPA: 1.2/1 (~ 9.7/10)
National and Kapodistrian University of Athens <i>BSc in Mathematics</i>	Athens, Greece GPA: 9.6/10

RESEARCH

- [5] **Dimitris Oikonomou** and Nicolas Loizou. *Adaptive Sharpness-Aware Minimization with a Polyak-type Step Size: A Theory-Grounded Scheduler*. Under Submission.
- [4] **Dimitris Oikonomou** and Nicolas Loizou. *Safeguarded Stochastic Polyak Step-Sizes for Non-smooth Optimization: Robust Performance Without Small (Sub)Gradients*. arXiv preprint arXiv:2512.02342, 2025
- [3] Robert Gower, Guillaume Garrigos, Nicolas Loizou, **Dimitris Oikonomou**, Konstantin Mishchenko, and Fabian Schaipp. *Analysis of an Idealized Stochastic Polyak Method and its Application to Black-Box Model Distillation*. arXiv preprint arXiv:2504.01898, 2025
- [2] **Dimitris Oikonomou** and Nicolas Loizou. *Sharpness-Aware Minimization: General Analysis and Improved Rates*. In **ICLR 2025**
- [1] **Dimitris Oikonomou** and Nicolas Loizou. *Stochastic Polyak Step-sizes and Momentum: Convergence Guarantees and Practical Performance*. In **ICLR 2025**

HONORS & AWARDS

Mathematical Olympiads - International Mathematical Olympiad (IMO): <i>Bronze Medal</i> - Greek Mathematical Olympiad: <i>Silver Medal</i> - Greek Mathematical Olympiad: <i>Bronze Medal</i>	Santa Marta, Colombia, 2013 Athens, Greece, 2014 Athens, Greece, 2013
Scholarships - Hellenic Post Scholarship - DAAD Scholarship - JHU MINDS Fellowship	2015-2019 2019-2020 2024, 2025

TECHNICAL SKILLS

Programming Languages: Python, R, C/C++, MATLAB
Libraries: PyTorch, NumPy, scikit-learn, Pandas
Applications: OpenMP, CUDA, $\text{\LaTeX}$

TEACHING & PROFESSIONAL SERVICE

Teaching Assistant: <i>Computer Vision</i>	JHU, Fall 2024
Reviewer: IEEE Transactions on Signal Processing, ICML, NeurIPS, ICLR	

LANGUAGES

English (Fluent), **Greek** (Native), **French** (Basic)